



Prototyping, Specification, Architecture & Design

Ashesi IRB Application Manager

Kelvin K. Ahiakpor

CS443: Mobile App Development

Dr. David Sampah

April 21, 2026

1 Application Overview & Architecture

Figures 1 and 2 show the high level, 3-tier system architecture. The presentation layer is mainly flutter-based for the IRB committee but there is also a web version mainly for students to submit applications or view their status. In the application layer, system logic is processed by the Supabase services to handle authentication, API requests, mailbox polling and integrations with external services including Microsoft Graph API, Ashesi Student API, notification and the SMS Gateway. The data layer uses Supabase's PostgreSQL and Storage.

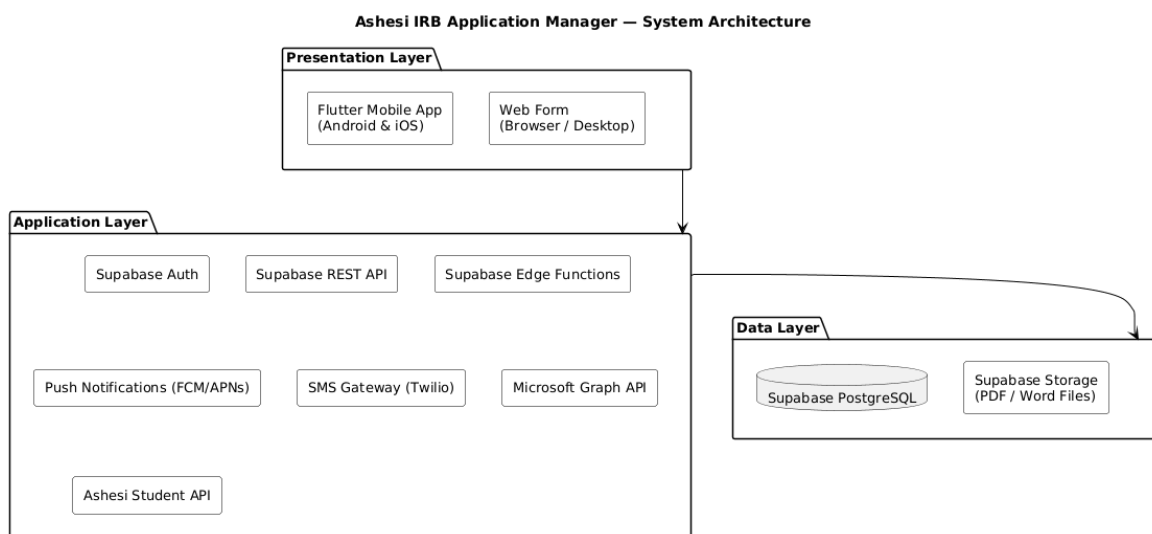


Figure 1 – Simple system architecture for the IRB Application Manager

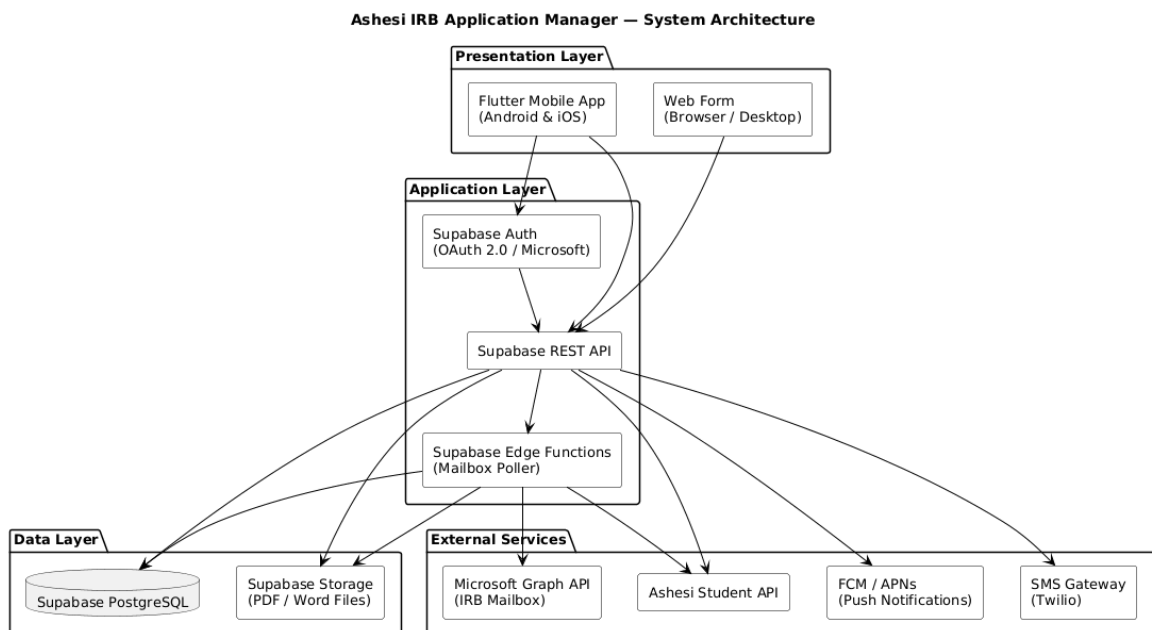


Figure 2 – Detailed system architecture for the IRB Application Manager

2 Use Case Diagram

Figure 3 shows the system's use case diagram. It shows how admins, reviewers and students use the application. Admins can oversee the platform. Reviewers manage applications reviewing documents and updating statuses. Students can check their application status.

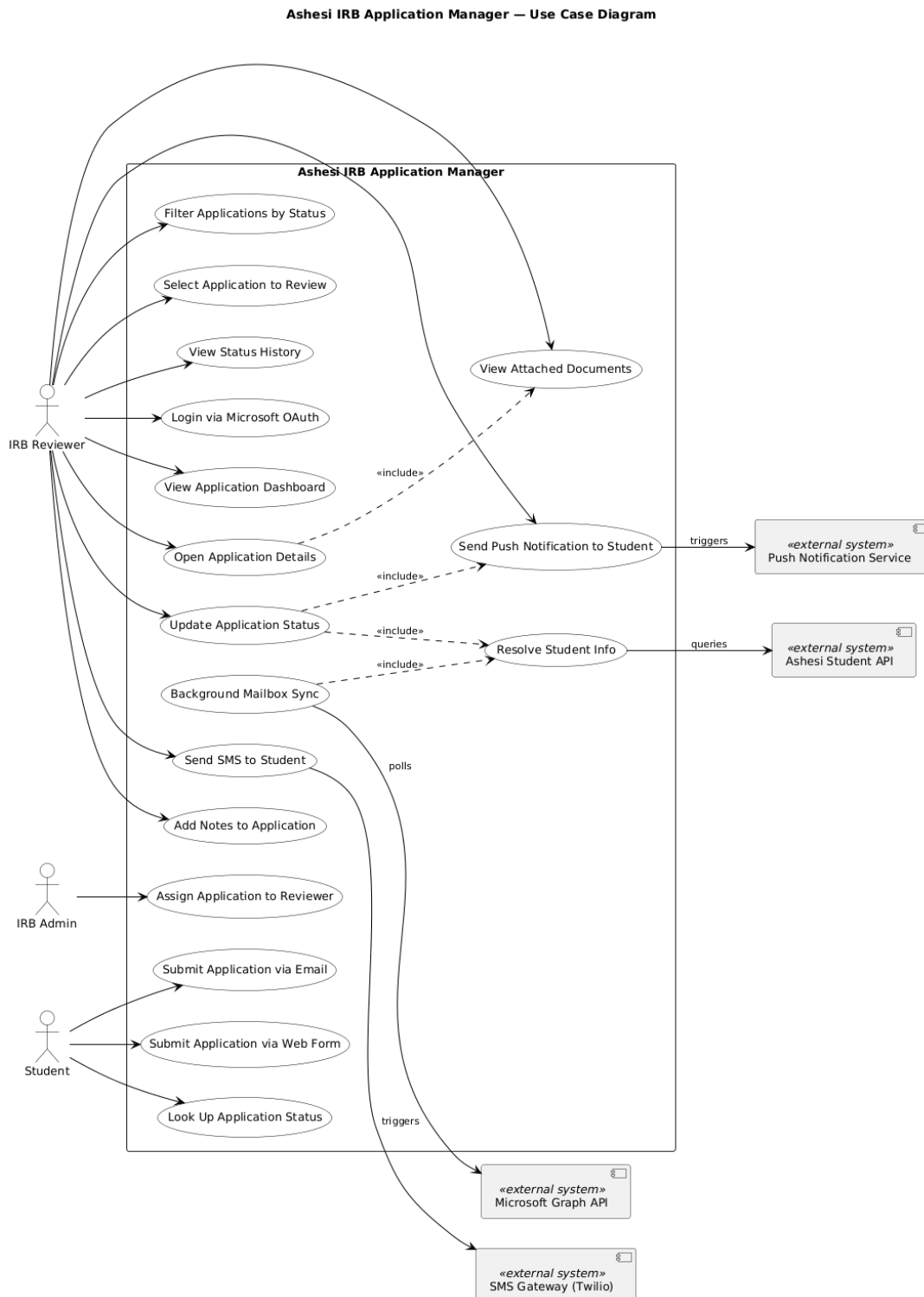


Figure 3 – Use case diagram for the IRB Application Manager

3 Entity Relationship Diagram

The application uses a relational database in its backend. The **reviewers** table stores information about IRB committee members who will review applications. The **applications** table is currently non exhaustive and will store all IRB submission details including student information, submission method, status, and the reviewer assigned to the application.

The remaining tables are support tables. The **attachments** table stores metadata for uploaded files such as PDFs and Word documents linked to each application. The **status_history** table records every status change made to an application, including the reviewer responsible and any notes added. The **notifications** table logs messages sent to applicants through SMS or push notifications.

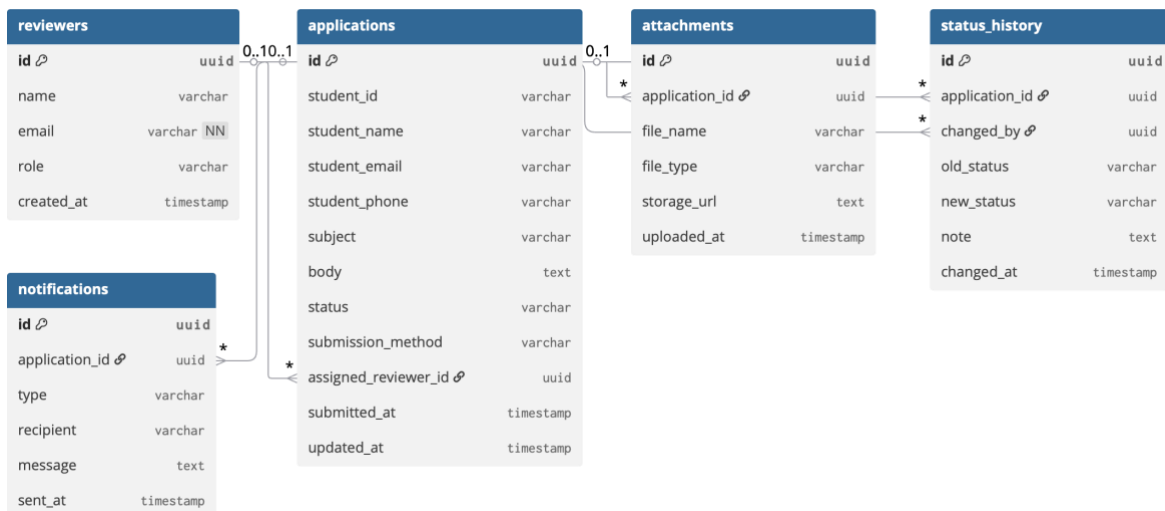


Figure 4 – Entity Relationship Diagram for IRB Application Manager

4 Prototype

The UI interfaces are shown in light and dark mode for the 9 screens in the application: Splash Screen, Login, Reviewer Dashboard, Application Detail, Document Viewer, Update Status Sheet, Status History, Web Submission Form, Student Status Lookup.

Splash Screen

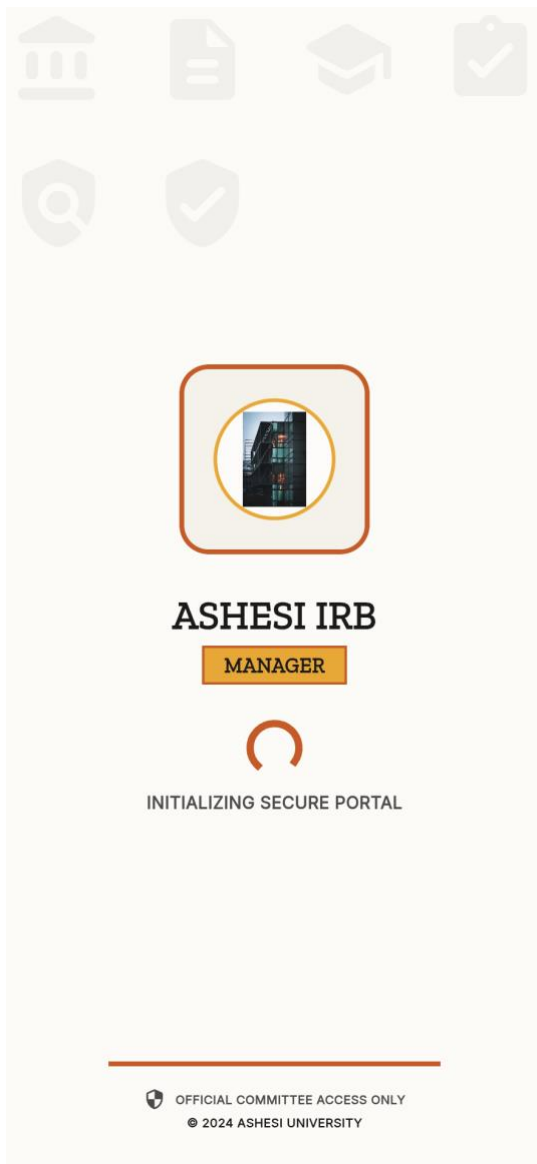


Figure 5 – Splash screen light mode

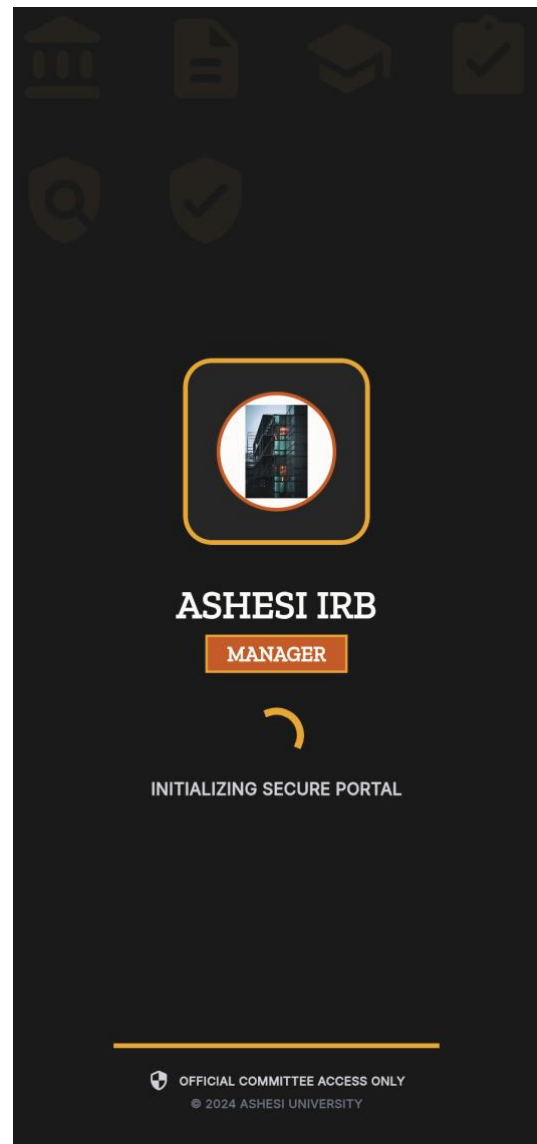


Figure 6 – Splash screen dark mode

Login Screen

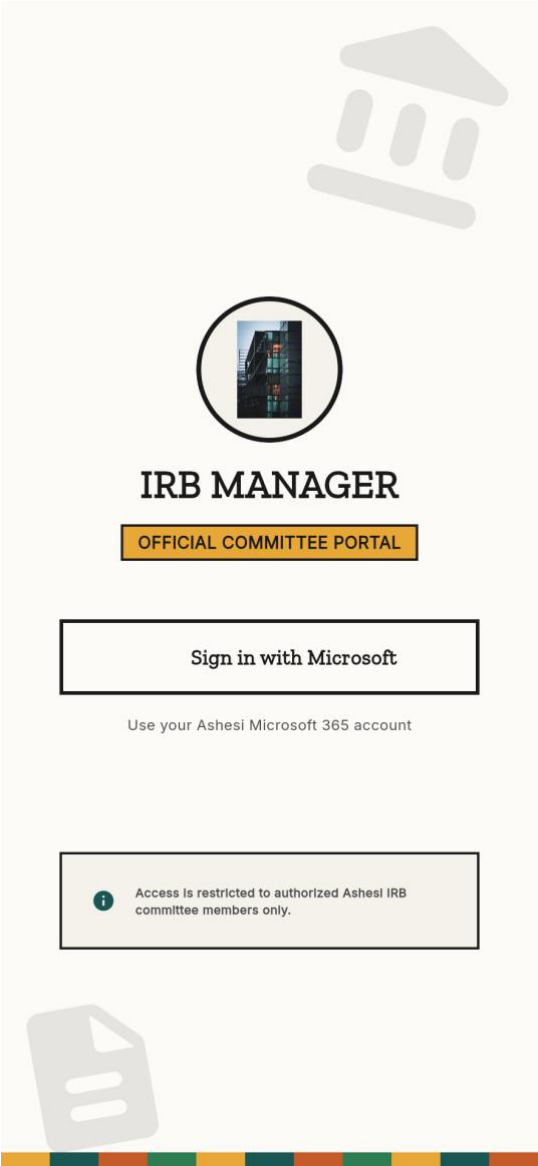


Figure 7 – Login screen light mode

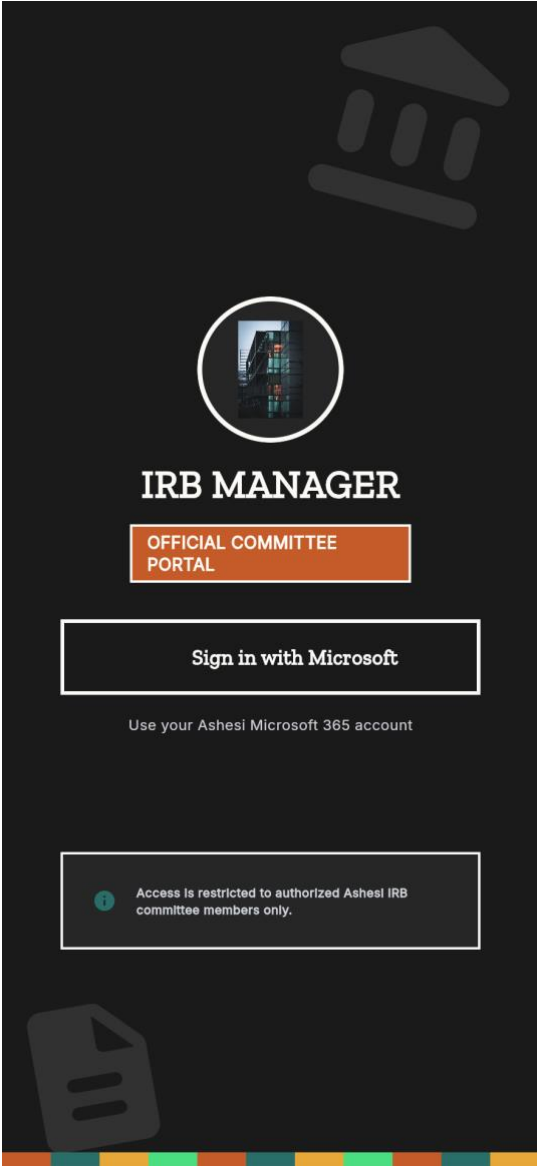


Figure 8 – Login screen dark mode

Reviewer Dashboard

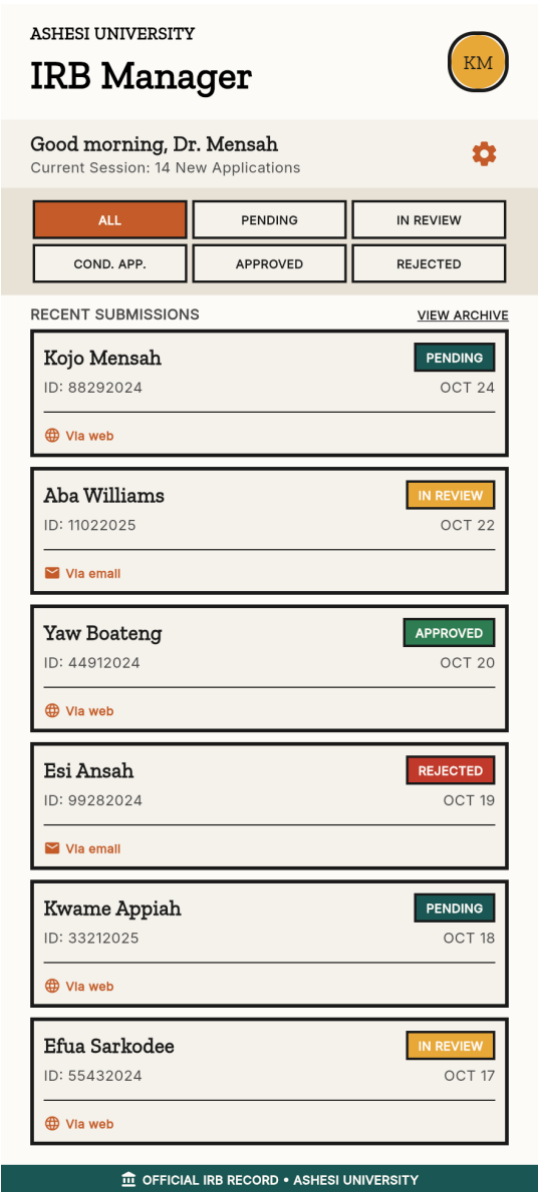


Figure 9 – Reviewer dashboard screen light mode

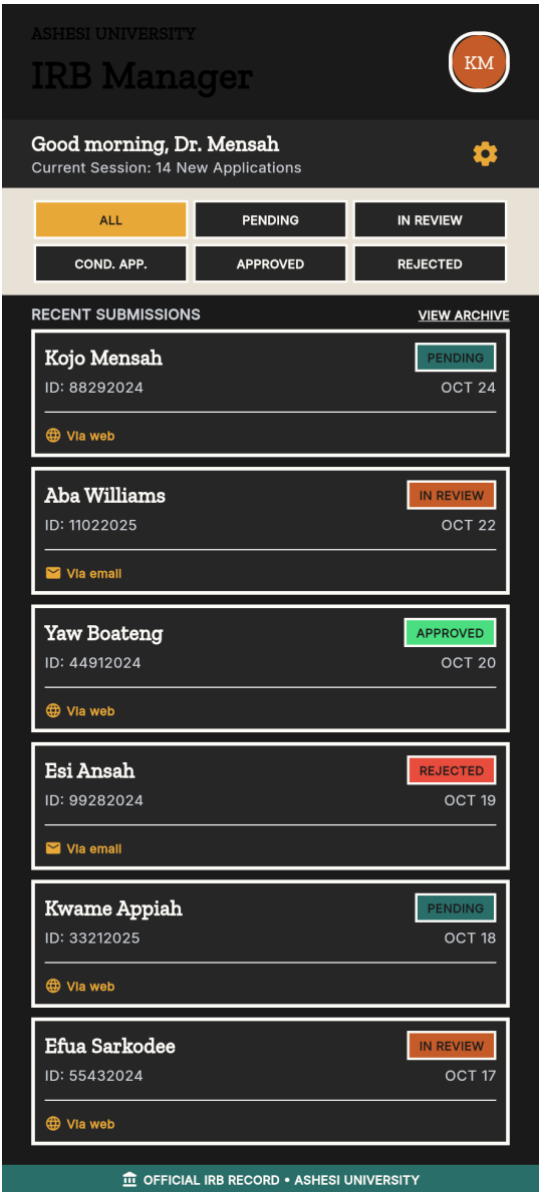


Figure 10 – Reviewer dashboard screen dark mode

Student Application Detail

←

APPLICATION DETAIL

⋮

Kojo Mensah

APPLICANT DETAILS

STUDENT ID

88292024

PHONE

+233 54 000 1234

EMAIL ADDRESS

kojo.mensah@ashesl.edu.gh

RESEARCH PROPOSAL

Impact of FinTech on Rural Trade

This study aims to investigate how mobile money adoption influences the trading patterns of small-scale farmers in the Eastern Region of Ghana. We will employ a mixed-methods approach, combining surveys with 200 participants and semi-structured interviews with local market leaders.

Submitted on Oct 24, 2023 via Web Form

ATTACHMENTS (3)

DOWNLOAD ALL

PDF

IRB_Application_Form.pdf

1.2 MB

📎

▼

DOCX

Consent_Letter_Template.docx

450 KB

📎

▼

Document Preview Placeholder

PDF

Survey_Questionnaire.pdf

890 KB

📎

▼

UPDATE STATUS

🔄

Figure 11 – Application detail screen light mode

←

APPLICATION DETAIL

⋮

Kojo Mensah

APPLICANT DETAILS

STUDENT ID

88292024

PHONE

+233 54 000 1234

EMAIL ADDRESS

kojo.mensah@ashesl.edu.gh

RESEARCH PROPOSAL

Impact of FinTech on Rural Trade

This study aims to investigate how mobile money adoption influences the trading patterns of small-scale farmers in the Eastern Region of Ghana. We will employ a mixed-methods approach, combining surveys with 200 participants and semi-structured interviews with local market leaders.

Submitted on Oct 24, 2023 via Web Form

ATTACHMENTS (3)

DOWNLOAD ALL

PDF

IRB_Application_Form.pdf

1.2 MB

📎

▼

DOCX

Consent_Letter_Template.docx

450 KB

📎

▼

Document Preview Placeholder

PDF

Survey_Questionnaire.pdf

890 KB

📎

▼

UPDATE STATUS

🔄

Figure 12 – Application detail screen dark mode

Document Viewer



Figure 13 – Document viewer light mode

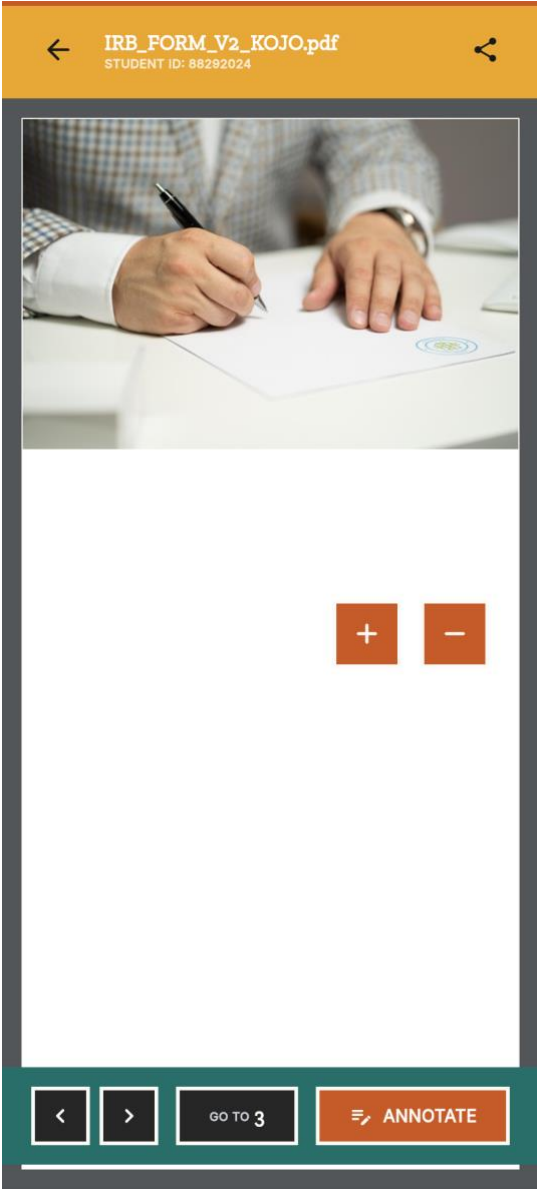


Figure 14 – Document viewer dark mode

Update Status

←

APPLICATION DETAIL

⋮

Kojo Mensah

ID: 88292024

PENDING

DECISION PORTAL

UPDATE STATUS

☒ SELECT OFFICIAL STATUS

☐ UNDER REVIEW

☒ CONDITIONALLY APPROVED

☐ APPROVED

☐ REJECTED

REVIEWER NOTES

Enter feedback for the student...

NOTIFY APPLICANT

Send automated alert to student

☒ EMAIL

☐ SMS

☒ CONFIRM STATUS UPDATE

☐ DISCARD CHANGES

Figure 15 – Update status screen light mode

←

APPLICATION DETAIL

⋮

Kojo Mensah

ID: 88292024

PENDING

DECISION PORTAL

UPDATE STATUS

☒ SELECT OFFICIAL STATUS

☐ UNDER REVIEW

☒ CONDITIONALLY APPROVED

☐ APPROVED

☐ REJECTED

REVIEWER NOTES

Enter feedback for the student...

NOTIFY APPLICANT

Send automated alert to student

☐ EMAIL

☒ SMS

☒ CONFIRM STATUS UPDATE

☐ DISCARD CHANGES

Figure 16 – Update status screen dark mode

Status History

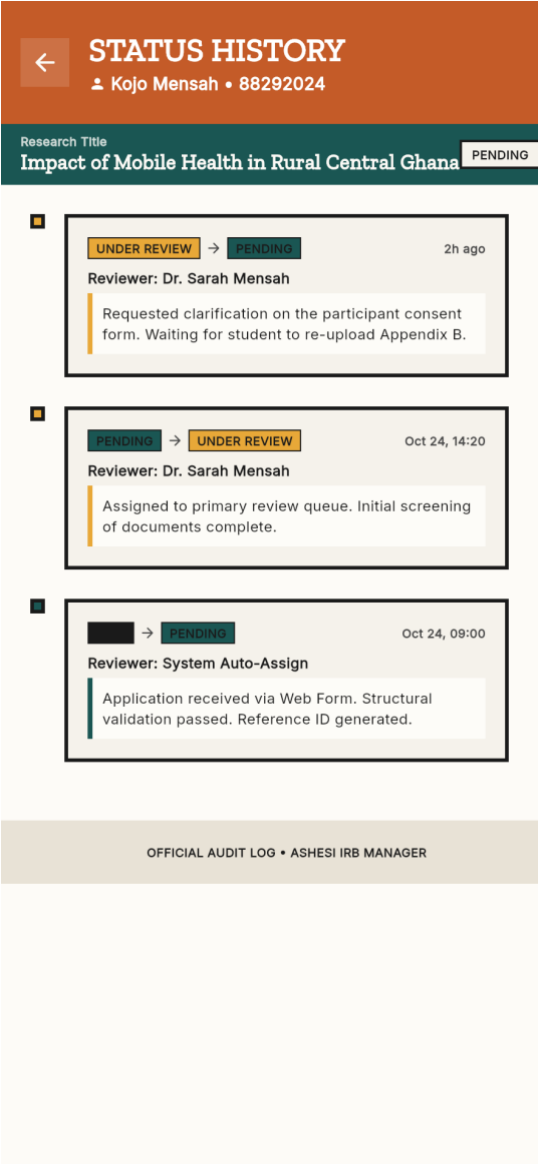


Figure 17 – Status history screen light mode



Figure 18 – Status history screen dark mode

Web Submission Form

ASHESI UNIVERSITY

IRB SUBMISSION

PERSONAL INFO

Full Name

Enter your legal name

Student ID

e.g. 88292024

Email

ashesi.edu.gh

Phone

024 XXX XXXX

RESEARCH DETAILS

Research Title

The full title of your study

Study Description

Briefly describe your research goals...

Recruitment Plan

How will you find participants?

ATTACHMENTS

Tap to upload IRB forms

PDF or DOCX (Max 10MB)

IRB_Form_Initial_v1.pdf

Consent_Letter_Template.docx

I confirm that all information provided is accurate and I have attached all required ethical clearance documents.

SUBMIT APPLICATION

OFFICIAL ASHESI IRB SYSTEM

For status checks, use the Status Lookup portal.

Figure 19 – Web submission form screen light mode

ASHESI UNIVERSITY

IRB SUBMISSION

PERSONAL INFO

Full Name

Enter your legal name

Student ID

e.g. 88292024

Email

ashesi.edu.gh

Phone

024 XXX XXXX

RESEARCH DETAILS

Research Title

The full title of your study

Study Description

Briefly describe your research goals...

Recruitment Plan

How will you find participants?

ATTACHMENTS

Tap to upload IRB forms

IRB_Form_Initial_v1.pdf

Consent_Letter_Template.docx

SUBMIT APPLICATION

OFFICIAL ASHESI IRB SYSTEM

For status checks, use the Status Lookup portal.

Figure 20 – Web submission form screen dark mode

Student Status Lookup

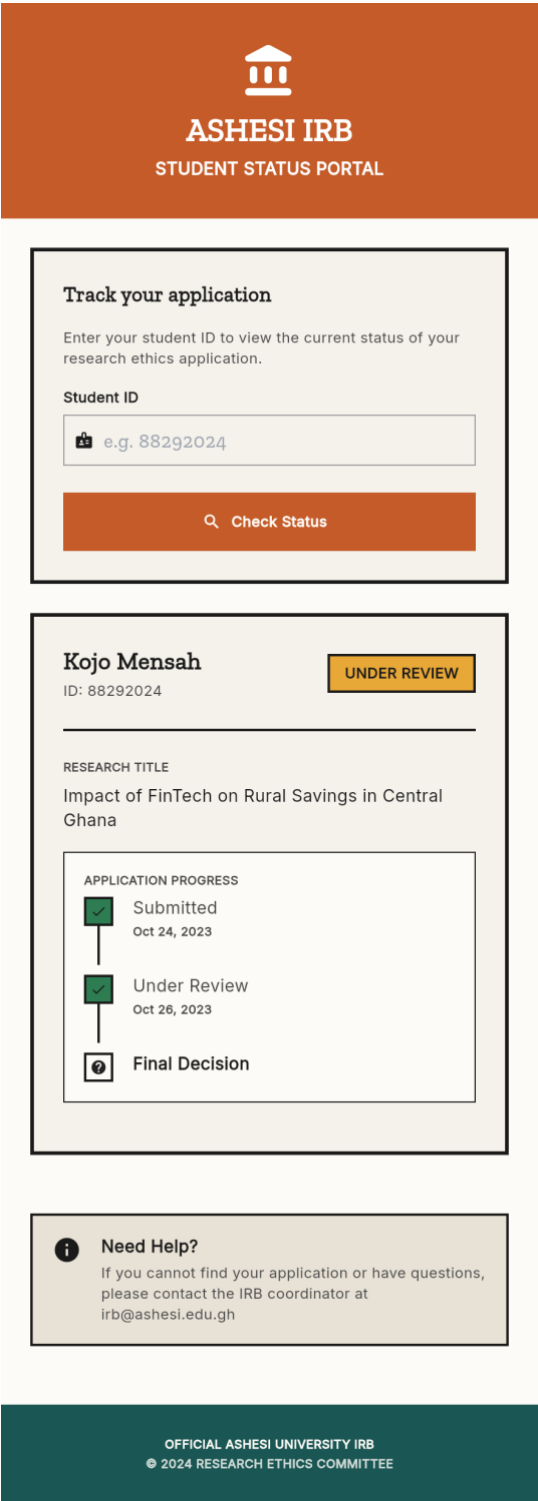


Figure 21 – Student status lookup screen light mode

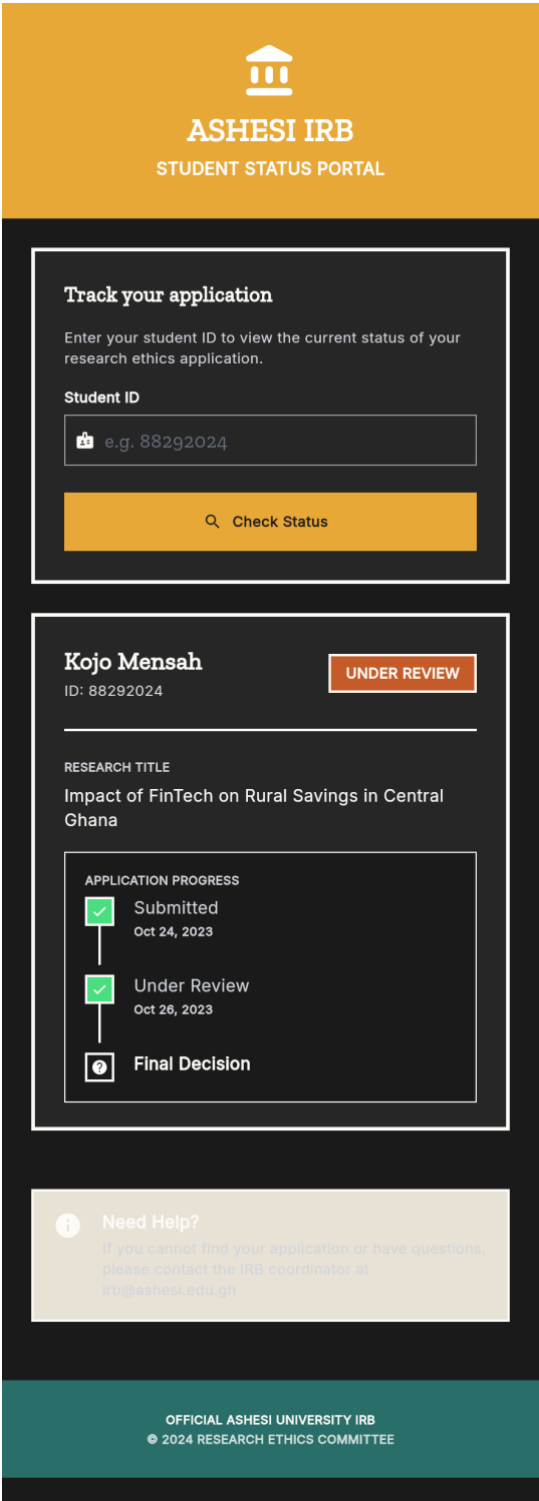


Figure 22 – Student status lookup screen dark mode